



Galaxy Classification

Using Convolutional Neural Networks

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1. Problem Definition

Galaxy morphology classification is the process of categorizing galaxies based on their visual structure. In this project, a Convolutional Neural Network (CNN) is trained to classify galaxy images into three major morphological types: **Spiral, Elliptical, and Irregular.**

Spiral galaxies have a rotating disk with prominent spiral arms. Elliptical galaxies appear as smooth, featureless ellipsoids. Irregular galaxies have no regular structure, often resulting from gravitational interactions or mergers.

The goal is to develop an automated image classifier that reduces the need for manual expert labeling of large-scale astronomical surveys.

2. Applications

- **Large-Scale Sky Surveys:** Missions like SDSS will image hundreds of millions of galaxies. Manual classification is infeasible; CNN-based classifiers are essential.
- **Cosmological Research:** Galaxy morphology is linked to galaxy formation and evolution physics.
- **Dark Matter Studies:** Spatial distribution of galaxy types helps constrain cosmological models.
- **Citizen Science Support:** Tools like Galaxy Zoo use crowd-sourced labels; CNNs can pre-screen images to reduce human workload.
- **Real-Time Telescope Pipelines:** Modern observatories need fast on-the-fly classification to prioritize follow-up observations.

3. Challenges

3.1 Class Imbalance The original dataset contains far more Spiral images than Elliptical or Irregular ones. Without correction, the model learns to favor the majority class. This was resolved by sampling exactly 1,000 images per class, creating a perfectly balanced dataset.

3.2 Visual Similarity Between Classes Irregular galaxies have no defining shape and can visually resemble disturbed spiral galaxies. This ambiguity makes Irregular the hardest class to classify, confirmed by our confusion matrix results.

3.3 Overfitting on Small Datasets With only 2,167 training images, a deep CNN tends to memorize training data rather than learn generalizable features. This was addressed using Dropout layers after each convolutional block and Data Augmentation.

3.4 Resolution Trade-off Raw images were 256×256 pixels, resized to 64×64 to reduce computational cost. Fine structural details such as thin spiral arms may be lost in this downsampling.

4. Methodology

4.1 Dataset Galaxy10 DECals dataset — 17,736 images from the DECam Legacy Survey. Three classes were selected and relabeled:

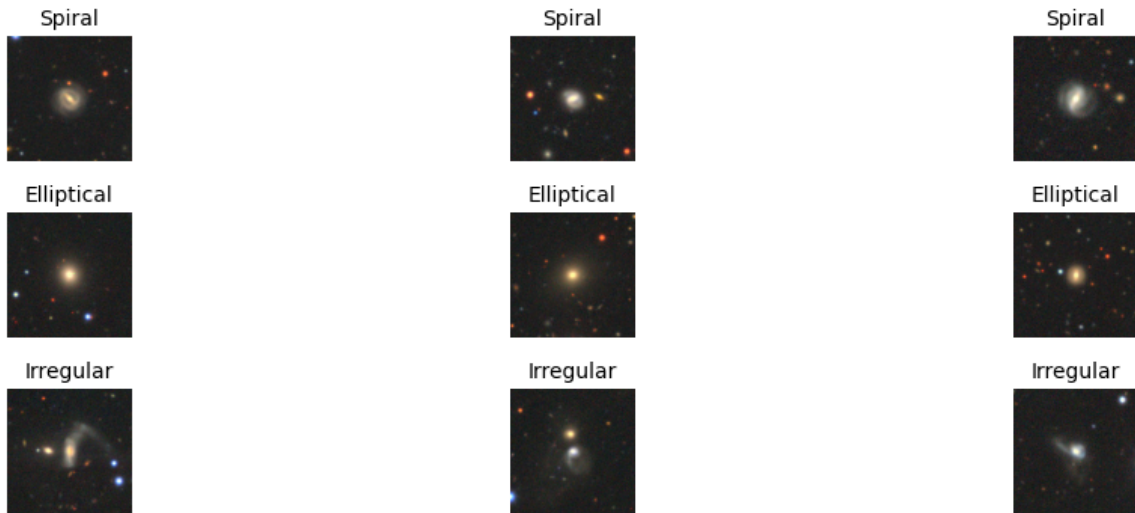
<i>class</i>	<i>Original Galaxy10 Labels</i>	<i>Samples Used</i>
<i>Spiral (0)</i>	5, 6, 7	1000
<i>Elliptical (1)</i>	2	1000
<i>Irregular (2)</i>	0	1000

Total: 3,000 images — perfectly balanced.

4.2 Preprocessing

- Normalization: Pixel values scaled from $[0, 255]$ to $[0.0, 1.0]$
- Resizing: $256 \times 256 \rightarrow 64 \times 64$ pixels
- One-Hot Encoding: Labels converted to vectors (Spiral = $[1, 0, 0]$)
-
- Data Split: 70% train / 15% validation / 15% test (stratified)

Galaxy sample image:



4.3 Data Augmentation Applied to training images via Keras ImageDataGenerator:

- Random rotation up to 30 degrees
- Horizontal and vertical flipping
- Zoom variation $\pm 10\%$
- Horizontal and vertical shifts up to 10%

4.4 CNN Architecture

<i>Layer</i>	<i>Output Shape</i>	<i>Parameters</i>
<i>Conv2D (32 filters, 3×3) + ReLU</i>	64×64×32	896
<i>BatchNormalization</i>	64×64×32	128
<i>MaxPooling2D</i>	32×32×32	0
<i>Dropout (0.25)</i>	32×32×32	0
<i>Conv2D (64 filters, 3×3) + ReLU</i>	32×32×64	18,496
<i>BatchNormalization</i>	32×32×64	256
<i>MaxPooling2D</i>	16×16×64	0
<i>Dropout (0.25)</i>	16×16×64	0
<i>Conv2D (128 filters, 3×3) + ReLU</i>	16×16×128	73,856
<i>BatchNormalization</i>	16×16×128	512
<i>MaxPooling2D</i>	8×8×128	0

Model: "sequential"		
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 64, 64, 32)	896
batch_normalization (BatchNormalization)	(None, 64, 64, 32)	128
max_pooling2d (MaxPooling2D)	(None, 32, 32, 32)	0
dropout (Dropout)	(None, 32, 32, 32)	0
conv2d_1 (Conv2D)	(None, 32, 32, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 32, 32, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 64)	0
dropout_1 (Dropout)	(None, 16, 16, 64)	0
conv2d_2 (Conv2D)	(None, 16, 16, 128)	73,856
batch_normalization_2 (BatchNormalization)	(None, 16, 16, 128)	512
max_pooling2d_2 (MaxPooling2D)	(None, 8, 8, 128)	0
dropout_2 (Dropout)	(None, 8, 8, 128)	0
flatten (Flatten)	(None, 8192)	0
dense (Dense)	(None, 128)	1,048,704
dropout_3 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 3)	387
Total params: 3,428,811 (13.08 MB)		
Trainable params: 1,142,787 (4.36 MB)		
Non-trainable params: 448 (1.75 KB)		
Optimizer params: 2,285,576 (8.72 MB)		

4.5 Training Configuration

- Optimizer: Adam
- Loss Function: Categorical Cross-Entropy
- Batch Size: 32
- Max Epochs: 40
- Early Stopping: patience=8, restore best weights
- Model Checkpoint: saved best model on validation loss

5. Evaluation Metrics

5.1 Accuracy Proportion of correctly classified samples out of all samples.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

5.2 Precision Of all samples predicted as class X, how many actually belong to class X. High precision = fewer false positives.

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

5.3 Recall (Sensitivity) Of all actual class X samples, how many were correctly identified. High recall = fewer missed detections.

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

5.4 F1-Score Harmonic mean of Precision and Recall. Provides a single balanced metric.

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

5.5 Confusion Matrix A table showing correct vs. incorrect predictions for each class pair. Diagonal = correct; off-diagonal = misclassifications.

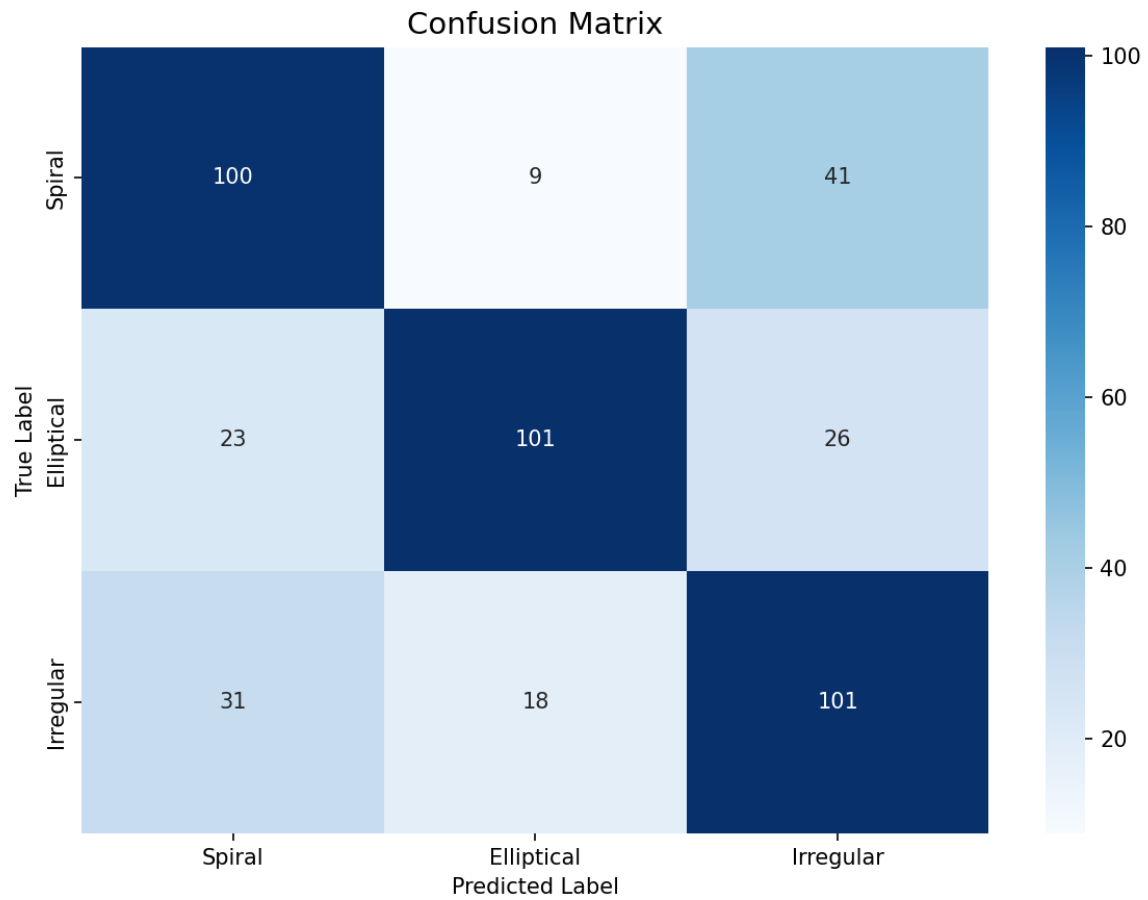
5.6 Cross-Entropy Loss Quantifies the difference between predicted probability distributions and true labels. Lower is better.

6. Results

6.1 Training Summary

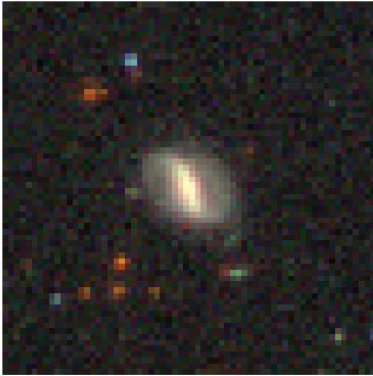
- Training stopped at Epoch 26 via Early Stopping
- Best validation accuracy: **71.28%**

6.2 Confusion Matrix (450 test images, 150 per class)



Sample Predictions (Green=Correct, Red=Wrong)

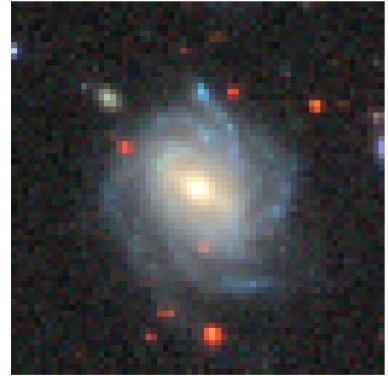
True: Spiral
Pred: Spiral



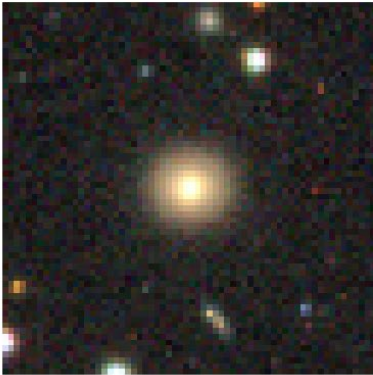
True: Spiral
Pred: Irregular



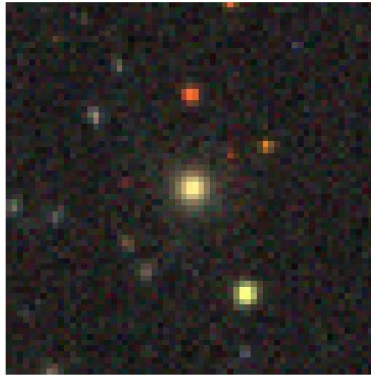
True: Spiral
Pred: Spiral



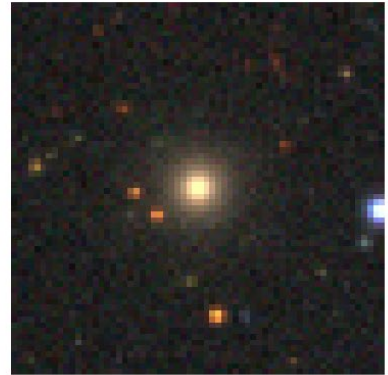
True: Elliptical
Pred: Irregular



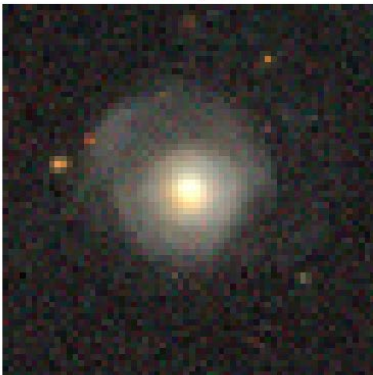
True: Elliptical
Pred: Elliptical



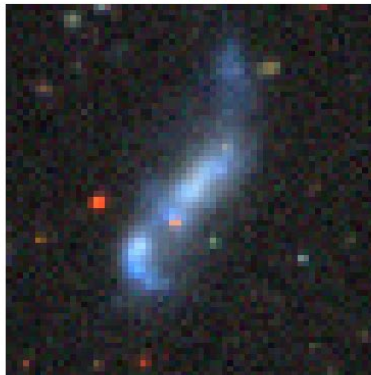
True: Elliptical
Pred: Elliptical



True: Irregular
Pred: Spiral



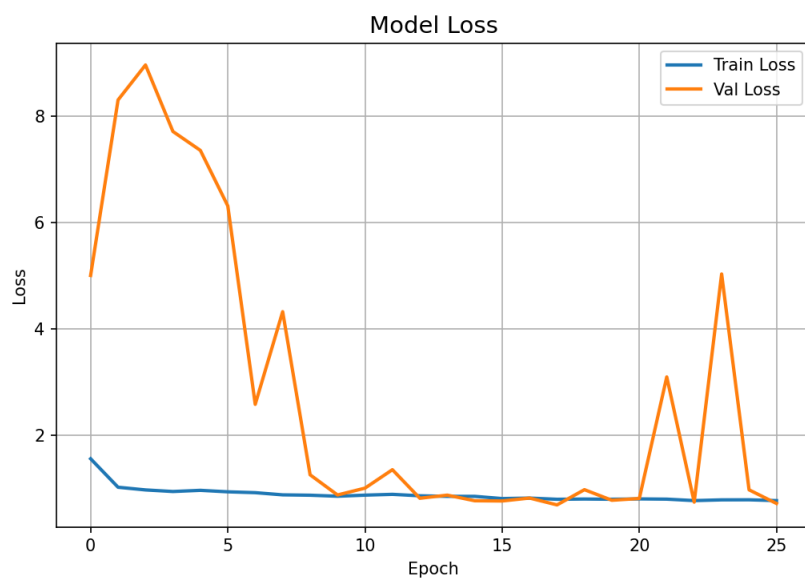
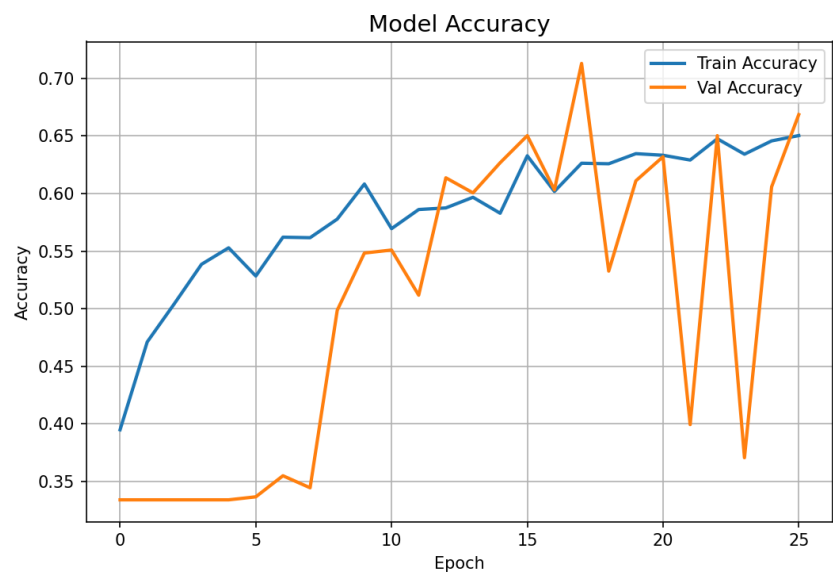
True: Irregular
Pred: Irregular



True: Irregular
Pred: Irregular



6.3 Per-Class Metrics



<i>class</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
<i>Spiral</i>	0.65	0.67	0.66
<i>Elliptical</i>	0.79	0.67	0.73
<i>Irregular</i>	0.60	0.67	0.64
<i>Overall Accuracy</i>			67.11%

6.4 Key Observations

- Elliptical had the highest precision (0.79), while Irregular had the lowest (0.60), indicating the model still confuses Irregular with other classes more often.
- Most common error: **Spiral galaxies misclassified as Irregular** (41 cases) — expected due to visual ambiguity between disturbed spiral structures and irregular morphologies.
- Data augmentation successfully prevented severe overfitting

7. Conclusion

This project demonstrated the feasibility of using a custom CNN to classify galaxy morphologies. With a balanced dataset of 3,000 images and regularization techniques (Dropout, BatchNorm, Augmentation), the model achieved 67.11% test accuracy.

Future improvements could include transfer learning with pretrained architectures (ResNet, VGG), larger dataset, or higher input resolution.

References

- Leung, H. W. & Bovy, J. (2019). Galaxy10 DECals dataset.